

Infrared Frame Evidence

Decode known pulse evidence without transmitting

An owned receiver and visible cadence preserve the evidence path.

Lesson	025 — component	Time	150–210 minutes
Prerequisites	Lessons 001–024	Board	Arduino Mega 2560
Primary evidence	D45 / TP-LED	Status	host verified; bench open
Safety boundary	E1, identified receiver and owned NEC remote; receive only		

1 Mission

Preserve the evidence chain:

capture → frame → validity → canonical record → visible cadence.

The Mega exercise captures an identified active-low receiver while an owned ordinary kit remote supplies known harmless NEC buttons. Deterministic host tests supply valid, unknown, timing, integrity, repeat, overflow, and record fixtures. Nothing emits infrared, exports raw pulses, maps a command to an action, or replays captured evidence.

Evidence before meaning. A numeric address or command is a decoded field, not an instruction such as “unlock”, “open”, or “launch”. Unknown or invalid evidence remains unknown or invalid.

1.1 Learning outcomes

- distinguish capture, frame, validity, and record;
- explain why invalid evidence is not an exceptional API failure;
- validate leader timing, bit timing, and inverse fields independently;
- encode a stable record into caller-owned bounded storage;
- observe decode classes without relying on Serial;
- explain pin, interrupt, snapshot, and acknowledgement ownership.

2 Parts, current, and wiring

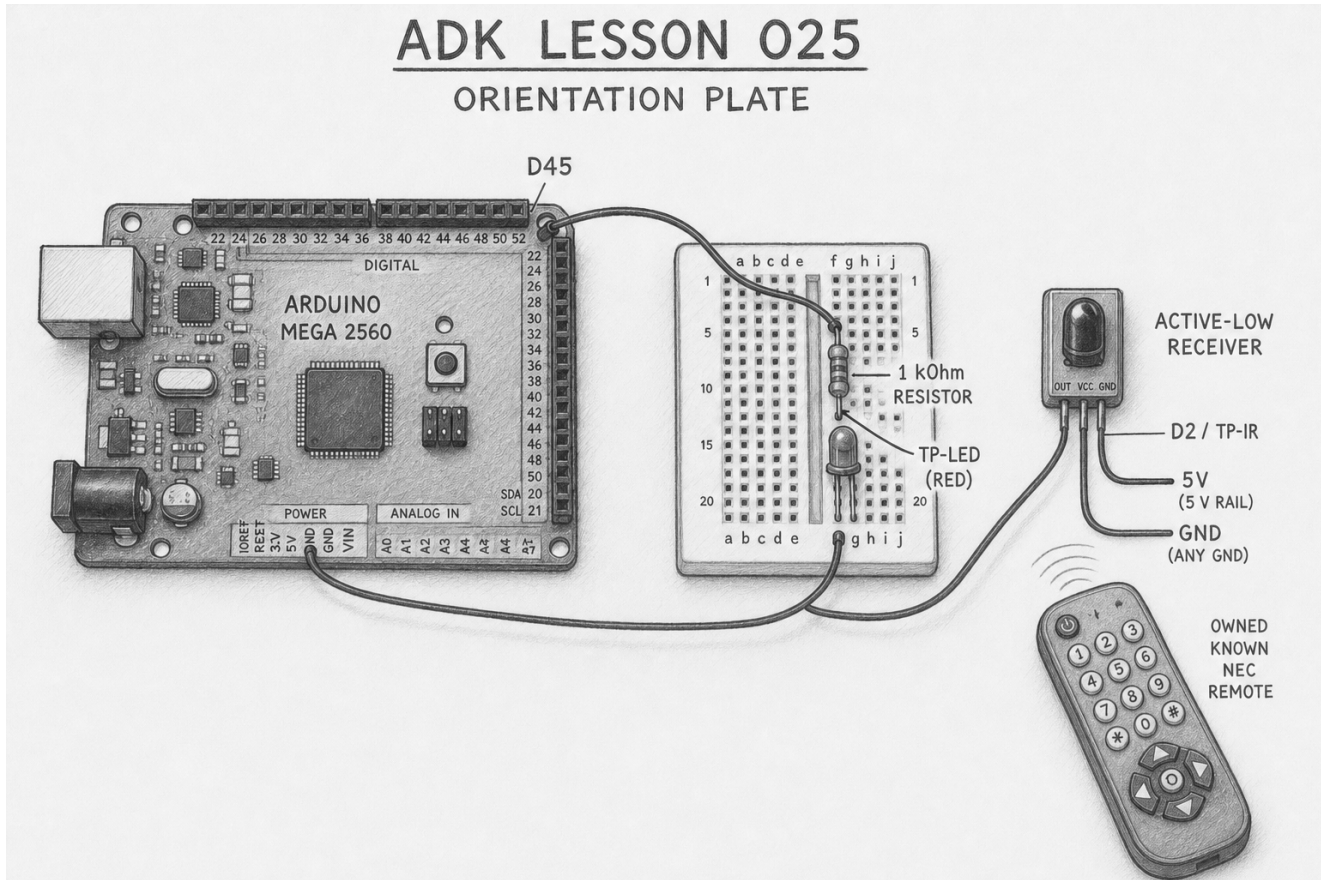
Part	Requirement
Arduino Mega 2560	USB-powered board
Evidence LED	One ordinary LED and one 1 kΩ resistor
Instrument	High-impedance logic probe or oscilloscope; meter for static state
Receiver	Exact active-low specimen with reviewed primary datasheet
Remote	Owned ordinary kit remote with known harmless NEC buttons

At an illustrative measured 5 V rail and 2 V LED drop,

$$I = \frac{5 \text{ V} - 2 \text{ V}}{1000 \text{ } \Omega} = 3 \text{ mA}.$$

Measure the actual rail, resistor, and LED drop before recording current. Remove USB power before wiring or attaching probe clips. Stop for unexpected heating, unstable power, a resistor mismatch, or voltage outside the rails.

Signal	Mega pin	Connection
Acquisition evidence	D13	built-in LED
Frame evidence / TP-LED	D45	D45 → 1 kΩ → LED anode; cathode → GND
Receiver evidence / TP-IR	D2	identified active-low receiver OUT
Receiver supply	5 V, GND	only if the exact primary datasheet permits it
Probe reference	GND	common reference for TP-LED and TP-IR



Type or operation	Contract
Pulse	one Mark or Space and bounded duration
PulseCapture	own pin/interrupt and publish stable bounded evidence
PulseFrame	immutable pulse span, sequence, and capture state
InfraredDecoder::decode()	validate evidence into an InfraredFrame
InfraredFrame	protocol, validity, numeric fields, and sequence
InfraredRecordEncoder::encode()	write one exact record into caller storage

The decoder recognizes the ordinary 8-bit-address NEC form used by the known fixture. It checks a leader, exactly 32 bits, a final mark, and both inverse bytes. Timing windows are inclusive. Noise before or after an otherwise valid sequence cannot be accepted as an embedded frame.

The first capture configuration accepts 100–10,000 μ s pulse durations and requires a 12,000 μ s frame gap. These are software gates, not optical range or receiver-performance claims. A frame stays stable until `acknowledge(sequence)` succeeds. An overrun cannot mutate it.

3.1 Status is not validity

An invalid pointer is an API error: `decode()` returns a non-OK **Status** and leaves the destination unchanged. A well-formed **PulseFrame** containing unknown, truncated, overflowed, or integrity-invalid evidence is different: decoding succeeds and the **validity** field carries that observation.

Invalid evidence keeps address and command zero. Code must branch on **validity** before using either numeric field.

3.2 Canonical record

```
IR1,<sequence>,<protocol>,<validity>,<address-hex>,<command-hex>\n
```

Decimal sequence spelling, uppercase names, eight uppercase hexadecimal digits, commas, and final newline are fixed. The encoder returns `Result<uint16_t>`; check `ok()` before reading `value()`. A short destination receives no partial record.

4 Narrative example

```
if (!observeInfrared (adk::MicrosecondTimePoint (micros ())))
{
    stopSafely ();
    return;
}

if (!showFrameEvidence (adk::TimePoint (millis ())))
{
    stopSafely ();
}
```

Objects appear in dependency order: runtime, capture adapter, owned capture, owned LEDs, pure decoder, then pure encoder. Setup acquires D2, its interrupt, and both evidence outputs before illuminating D13. The loop updates capture, decodes a stable frame, records it, acknowledges its sequence, and shows D45.

Host tests replay these deterministic fixture classes:

Fixture	Deliberate evidence	D45 result
Known NEC	valid leader, 32 bits, inverse fields, final mark	one 1 s pulse
Unknown	leader outside the accepted interval	two 150 ms pulses
Fault	timing, integrity, overflow, or record failure	four 100 ms pulses

Record output is optional supporting evidence. The LED cadence and TP-LED waveform are the primary observation path.

5 Predict, observe, interpret

5.1 Experiment A: acquisition

Predict. D13 illuminates only after D2, its interrupt, and both LED endpoints initialize.

Observe: _____.

Interpret. This supports acquisition. It does not establish capture timing, D45 cadence, or safe shutdown.

5.2 Experiment B: class cadence

Before uploading, identify the receiver from its marking and primary datasheet. Predict TP-IR idle high and an active-low frame from one known button. Attach both probe channels while unpowered.

Button/result	TP-IR frame	D45 count	D45 widths
Known NEC	_____	_____	_____
Known repeat	_____	_____	_____
Deliberate fault	_____	_____	_____

Agreement supports the path from light through capture and decode to visible output. It does not establish optical range or general compatibility.

5.3 Experiment C: stable record

Optionally run the CLI serial logger. Predict a record only after the owned remote's harmless numeric fixture fields have been reviewed. Observed bytes: _____.

Compare exact case, field order, widths, and newline. A log never substitutes for TP-IR and TP-LED.

5.4 Experiment D: safe state

Invoke the stop path. Predict both LEDs inactive and D45 released. Observe the LED, then measure the pin separately with the appropriate high-impedance instrument. A dark LED alone does not prove high impedance or claim release.

Recorded safe-state evidence: _____.

6 Diagnosis

Observation	Check next
D13 dark	D2 interrupt or LED claim conflict, initialization status, board power
D13 on; D45 dark	TP-IR activity, D45 polarity, resistor, LED orientation
No complete frame	receiver identity, idle gap, pulse limits, interrupt ownership
Valid becomes unknown	leader levels, durations, exact pulse count
Valid becomes fault	mark/space timing and inverse bytes
Serial record absent	encoder result and caller buffer; use D45 as primary evidence
D45 remains driven after stop	safe-state failure; remove USB and record deviation

7 Owned receive-only boundary

PulseCapture and MegaPulseCaptureIo own the D2 pin and interrupt. For the identified active-low demodulating receiver, LOW maps to Mark and HIGH maps to Space. The ISR records bounded timestamps only; decoding, records, and policy remain outside the ISR.

Before powering a module, record its exact marking and primary datasheet. Verify supply range, pinout, idle polarity, carrier band, output levels, and Mega compatibility. A three-pin module that resembles another receiver does not inherit its ratings.

Capture acceptance remains incomplete until host tests prove:

- pin and interrupt ownership with reverse-order rollback;
- frame-gap, minimum-pulse, and maximum-pulse validation;
- stable completed snapshots until matching acknowledgement;
- explicit overflow, timing-fault, overrun, and stale-acknowledgement results;
- shutdown during capture, interrupt disable, and resource reuse;
- timestamp-wrap behavior and deterministic replay.

Use only an owned ordinary kit remote and named harmless buttons. Do not export or replay unknown captures. No transmitter, jammer, brute-force mapping, credential action, or safety-critical command belongs in this lesson.

8 CLI and open bench record

```
make host-test
make arduino-Lesson025InfraredEvidence
make firmware-size EXAMPLE=Lesson025InfraredEvidence
make upload EXAMPLE=Lesson025InfraredEvidence PORT=/dev/ttyACM0
make serial-log PORT=/dev/ttyACM0
make lessons
make lessons-check
make site-check
```

Software: commit _____; Arduino CLI _____; AVR core _____; flash _____; RAM _____.

Bench acceptance—open until measured

- ☐ Exact Mega, receiver, remote, LED, and resistor recorded.

- ☐ Receiver datasheet, pinout, supply, and idle polarity recorded.
- ☐ Unpowered D2/D45 wiring and polarity checked.
- ☐ D13 acquisition evidence observed.
- ☐ TP-IR active-low pulse train measured for known buttons.
- ☐ One-, two-, and four-pulse D45 patterns measured.
- ☐ TP-LED pulse widths recorded with instrument and reference.
- ☐ Canonical records compared byte for byte, if Serial used.
- ☐ Reset, shutdown, USB removal, and claim reuse observed.
- ☐ D45 safe state checked separately from LED appearance.

Supply _____; instrument _____; date _____; reviewer _____. Deviations: _____.

Current result: host verified; hardware acceptance open. Deterministic fixtures, compilation, a PDF, and blank worksheets are not physical verification.

9 Exercises and primary reading

1. Change one inverse byte while preserving valid timings. Predict the validity and cadence before running a host test.
2. Prove that a one-byte-short record buffer remains unchanged.
3. Explain why **UnknownProtocol** belongs in frame validity while a null pulse pointer belongs in **Status**.
4. Design a receiver overrun experiment that never transmits or exports the captured waveform.
5. Sketch how a matching sequence acknowledgement prevents a stale caller from clearing newer evidence.
 - HTML reference and corrections
 - PulseCapture public header and implementation
 - Mega capture adapter and capture lifecycle tests
 - InfraredDecoder public header and implementation
 - InfraredRecordEncoder public header and golden-vector tests
 - Arduino Mega 2560 Rev3 datasheet
 - Vishay TSOP382 receiver-family datasheet: example identification evidence, not a claim about an unmarked kit module
 - NEC protocol description: secondary timing reference for owned fixtures